

**NUTRIENT INTAKE, DIGESTIBILITY AND GROWTH PERFORMANCE OF YANKASA RAMS FED GRADED LEVELS OF SOYBEAN HUSK IN NORTH WESTERN NIGERIA**¹Jinjiri, A., ²Salisu, S., ²Auwal, Z.M., ¹Isa, M., ¹Sa'ad, S.K., ³Ali, M., & ²Nasiru, A.¹Rabiu Musa Kwankwaso College of Advanced and Remedial Studies, Tudun Wada Dankadai, Kano, Nigeria²Department of Animal Science, Bayero University, Kano³Departments of Agricultural Extension and Management, Binniyaminu Usman Polytechnic, Hadejia, Jigawa State*Corresponding author: jinjiri.aminu001@gmail.com +2348064980565**Keywords:**diets,
digestibility,
intake,
proximate
composition,
weight gain.**ABSTRACT**

This study was aimed to determine the effect of feeding diets containing graded levels of soybean husk on the nutrient intake, digestibility and growth performance of Yankasa rams. Twelve (12) Yankasa rams with an average weight of 20.75kg were selected from the Livestock Unit of the Department of Animal Science, Bayero University, Kano and were randomly assigned to three (3) dietary treatment groups with four animals per group in a randomized complete block design (RCBD). The treatment diets were; 0%, 15% and 30% soybean husk inclusion level. The result of the daily weight gain was not significantly ($P>0.05$) affected by the level of inclusion of the test materials. The average daily feed intake did not differ significantly ($P>0.05$) among the treatments, with values ranging from 770.75 to 821.00 g/day. However, the results of the nutrient digestibility revealed significant ($P<0.05$) differences in dry matter digestibility (DMD), crude fibre digestibility (CFD), nitrogen free extract digestibility (NFED) and neutral detergent fibre digestibility (NDFD) between treatment groups. It is concluded that Soybean husk can be fed for up to 30% inclusion level to Yankasa Rams without any harmful effect on growth. It is therefore recommended that further feeding trial be conducted with other breed of sheep during feed scarcity.

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INTRODUCTION

Sheep population in Nigeria is estimated to be 33.9 million and is primarily kept for meat production (FAO, 2008). Garba and Sunusi (2019) reported that sheep are found throughout Nigeria, but the highest population is found in Northern part where the three of the major breeds of sheep are found, the breeds are *Yankasa*, *Uda* and *Balami*. In Nigeria, sheep play a very significant role during cultural and religious festivals such as naming ceremony. Sheep production is the best way of enhancing meat supply and generates income to small holder farmers. Nevertheless, effective and efficient sheep production has been hindered by the cost of conventional feedstuffs such as soybean cake, cotton seed cake and fish meal needed to supplements low quality feed (Akinmutumi, 2004) and also the competition with humans and monogastric animals (Ukpabi & Abdu, 2009). There is a shortage of forage supply in the North-western region of Nigeria where the population of sheep is very high due to rainfall distribution pattern (Mohammed & Saleh,

2018). For effective production of animal protein, livestock farmers have embarked on intensive management system of rearing animal through local confinement (Ademosun, 1992).

Nigeria is the largest soya bean producer in Sub-saharan Africa (Kamara et al., 2023). Soya bean is a leguminous crop that provides food for humans, feed for animals and raw materials for industries (Abbey et al., 2001). It is an excellent source of protein (35 – 40%). Soybean and its products are cheap and the richest source of protein for the whole animal industry especially to accomplish the protein requirement (Ali et al., 2020). The soybean used in animals is very common due to its nutritional and functional possessions especially as meat extenders in animals (Belloque et al., 2002). The use of soybean husk is very common in the present generation due to its high nutritive value in ruminant and highly digestible fibre source that increases rumen micro flora flow and rumen digestion. Chellapandian (2019) reported that soybean husk contained 12.32% CP, 32.54% crude

fibre, 2.70% ether extract, 66.10% NDF and 49.72% ADF. Therefore, soybean husk can be considered a high fibre, moderate protein feed ingredient with useful energy value. Its fibrous nature and relatively good digestibility make it suitable for incorporation into ruminant diets. The objective of this study therefore was to determine the nutrients intake, digestibility and growth performance of Yankasa rams fed graded levels of soybean husk.

METHODOLOGY

Description of the study area

The experiment was conducted at the Livestock Teaching and Research Farm, Department of Animal Science, Bayero University, Kano which is located along Gwarzo Road in Rimin Gata, Kano State, north western Nigeria. The campus is situated approximately at 11. 972⁰ N latitude and 8.425⁰ E longitudes. The university is located within Sudan/Sudanian Savanna ecological zone of Nigeria, which is characterized by a distinct wet and dry season (BUK, 2015).

The climate of the study area is generally hot, with a relatively high temperatures occurring during most of the year. The rainy season usually extends from approximately May to October, while the dry season occurs from November to April (BUK, 2015).

Experimental animals

Twelve (12) Yankasa rams with an average weight of 20.75kg obtained from the Livestock Unit of the Department of Animal Science, Bayero University, Kano were used for the study.

Experimental diets

Soybean husk, maize bran and rice bran were obtained from the Department of Agronomy, Bayero University, Kano. Three diets were formulated containing cotton seed cake, wheat offal, maize bran, brewers dried grain, rice bran and common salt (Table 1). The experimental diets were designated as T1, T2 and T3 with 0%, 15% and 30% inclusion levels of soybean husk respectively.

Experimental design

Twelve (12) Yankasa rams were randomly assigned into three dietary treatment groups with four animals per group in a randomized complete block design (RCBD).

Management of the Experimental Animals

Prior to the commencement of the experiment, the animals were treated against ecto and endo parasites using ivermectin injection at 2ml per 10kg body weight and albendazole at 1ml per 50kg body weight. The animals were randomly allocated to three (3) groups of four (4) animals each. Adjustment period of one week was allowed for the animals before the commencement of the data collection.

Table 1: Gross Composition of the Experimental Diets (%)

Ingredients (%)	Treatments		
	T1 (0%)	T2 (15%)	T3 (30%)
Soybean husk	0	15	30
Cotton seed cake	18	27	39
Maize bran	9	4	4
Brewers dried grain	31	24	18
Rice bran	14	9	4
Wheat offal	27	20	4
Salt	1	1	1
TOTAL	100	100	100
Estimated CP%	15.72	15.21	14.25
Estimated Energy	4290	4302	4260

Data Collection

Performance indices

The experimental animals were allocated to the three treatments groups comprising four (4) animals per treatment and housed individually. The experimental diets formulated were given to the animals every morning throughout the period of the experiment which lasted for eight weeks.

The animals were weighed at the beginning of the experiment and on weekly basis to determine average weight changes. Data on feed intake and body weight gain were evaluated. Feed intake was obtained by subtracting left over feeds from the feeds offered the previous morning. The live weight gains were determined by difference between final weight and Initial weight (Final – Initial).

The average daily weight gain was determined by dividing weight gain by the number of sheep and the number of days of the feeding trial.

Digestibility trial

At the end of the feeding trial which lasted for eight weeks, digestibility trial was conducted. The animals were harnessed with harness bags for adaptability to the bags. Faecal output was collected and weighed. It then mixed thoroughly and weighed, some portion were taken to the laboratory for analysis (AOAC, 2005).

Chemical Analysis

Each sample of the experimental diet was milled to pass through a 1.00mm screen using a telator cyclotee 1093. The milled samples were used for proximate analysis to determine Nitrogen (N) for use in crude protein determination ($N \times 6.25$), crude fibre (CF), ether extract (EE) and ash according to AOAC (2005). Neutral Detergent fibre and acid detergent fibre (ADF) were determined according to procedures outlined by Van Soest et al. (1991).

Statistical analysis

Data obtained were subjected to analysis of variance (ANOVA) using SPSS package

(1999).

RESULTS AND DISCUSSIONS

Proximate Composition and the Crude Fibre fraction of the Experimental Diets

The proximate composition and the crude fibre fraction of the experimental diets fed to the growing Yankasa rams are presented in Table 2. The values for dry matter content were 94.00%, 96.00% and 94.00% for T1, T2 and T3 respectively. These values were in harmony with 89.6% obtained by Ettu & Owuka (2015). The crude protein values obtained ranged from 15.16 – 16.72%. Therefore, the present study CP content indicated that, it is a good source of feed that can help proper function of rumen microbes and also meet the maintenance requirement of animals (Van Soest, 1994).

The crude fibre in the present diets varies. The result was 16.50, 15.91 and 17.06% for the inclusion levels of 0%, 15% and 30% respectively. The result of the present study was higher than the values 35.93 – 41.93% reported by Chana et al. (2024) for sheep fed graded level of soybean waste in a total mixed ration.

Table 2: Proximate Composition and Crude Fibre Fraction of the Experimental Diets (%)

Parameters (%)	Treatments		
	T1 (%)	T2 (15%)	T3 (30%)
Dry matter	94.00	96.00	94.00
Crude protein	16.72	15.84	15.16
Crude fibre	16.50	15.91	17.06
Ash	6.40	6.11	6.78
Ether extract	2.94	3.19	3.13
Nitrogen free extract	57.45	58.96	57.87
Acid detergent fibre	54.67	50.24	30.68
Neutral detergent fibre	60.26	57.97	51.07

Performance Characteristics of the Experimental Animals

Table 3 shows performance of the experimental animals fed the test diets. Comprising of the initial weight, final weight gain and the average daily weight gain of the Yankasa rams fed graded levels of Soybean husk. The average daily weight gain was not significantly ($P>0.05$) affected by the inclusion levels of the test materials. The average daily weight gain in this study was very higher in

15% inclusion level (188.10g/day) while 0% inclusion level had the lowest (142.6g/day) and these values were higher to those obtained (50 – 106 g/day) by Chana et al. (2024) in a study to determine the effect of soybean waste on the performance of sheep. The differences could be attributed to the breed, sex and environment of the sheep. The feed efficiency values revealed that T2 was the best (0.244) followed by T3 (0.206) and T1 was the least (0.018).

Table 3: Performance of Yankasa Rams Fed Graded Levels of Soybean Husk

Parameters	Treatments			S.E
	T1 (0%)	T2 (20%)	T3 (30%)	
Initial weight (kg)	20.25	20.75	21.25	2.02
Final weight (kg)	23.25	24.70	24.80	2.69
Weight gain (kg)	3.00	3.95	3.55	0.21
Average daily weight gain (g/day)	142.86	188.10	169.05	1.44
Feed efficiency	0.181	0.244	0.206	0.13
FCR	5.52	4.10	4.86	0.29

FCR= feed conversion ratio; S.E = Standard Error

Nutrients Intake

The nutrient intake of the Yankasa rams fed the graded levels of the Soybean husk is presented in Table 4. The average daily feed intake was not significant ($P>0.05$) among the treatments ranging from 770.75 – 821.00g/day. The average feed intake in the present study was higher than 610.83 – 671.2 g/day reported by Mijinyawa et al. (2024) for Yankasa rams. This difference might be due to variations in the proportion of the feed ingredients, age, sex and other management systems of the sheep. The dry matter intake (DMI) did not differ significantly ($P>0.05$) among the treatments.

However, 30% inclusion levels had the highest DMI at 771.75g/day while 15% inclusion level recorded the lowest at 739.75g/day.

The crude protein intake did not significantly differ across the inclusion levels ($P>0.05$). T1 recorded the highest CP intake (122.30g/day) followed by T2 (116.85g/day) and the least is 116. 60 g/day for T3.This variation could be due to the presence of some anti nutritional factors in the soybean husk which might have interfered with the intake. Dry matter intake did not differed significantly ($P>0.05$), although slight differences existed and highest value was at 30% inclusion level.

Table 4: Nutrients Intake of Yankasa Rams Fed Graded Levels of Soybean Husk

Parameters	Treatments			S.E
	T1(0%)	T2(15%)	T3(30%)	
Average daily feed intake (g/d ay)	788.75	770.75	821.00	47.70
Daily Dry Matter Intake (g/day)	741.50	739.75	771.75	45.00
Daily Crude Protein Intake (g/day)	122.30	116.85	116.60	3.26
Daily Crude Fibre Intake (g/day)	125.80	118.30	132.55	16.22
Daily Ether Extract Intake (g/day)	21.50	23.50	24.00	1.70
Daily Nitrogen Free Extract Intake (g/day)	423.40	435.80	446.00	19.50
Daily Ash Intake (g/day)	48.50	45.30	52.60	5.55
Daily Acid Detergent Fibre Intake (g/day)	404.10 ^a	371.70 ^a	237.40 ^b	21.90
Daily Neutral Detergent Fibre Intake (g/day)	449.00	429.00	395.00	31.80

S.E = Standard Error

Digestibility

Table 5 shows the result of the digestibility study. According to FAO (1991) digestibility of feed can be classified as high when it is (greater than 60), medium (40-60) and low (less than 40). Therefore, all the digestibility indices evaluated in the present study were high. Muhammad et al. (2016) observed similar trend when Uda rams were fed graded levels of *Parkia biglobosa*. The dry matter digestibility ranged from 71.35 (30%) to 75.26 (0%). However, 0% and 15% were statistically similar but differed ($P<0.05$) significantly from 30% inclusion level. The values in the present study were not in accordance with the

27.57 – 55.73% values obtained by Chana et al. (2024) who evaluated the nutrient digestibility of sheep fed graded levels of Soya bean waste in a total mixed ration. These differences could be due to interaction and associative effect with the animal and also the variations in the test materials (Preston & Leng, 1986; Waba & Bandawa, 2019). The crude protein digestibility of the inclusion levels ranged from 78.68% (T2) to 80.00% (T3). There was no significant difference ($P>0.05$) among the treatments of the Soybean husk.

The crude fibre digestibility ranged from 45.09% (T3) to 62.56% (T1). But, soybean

husk inclusion level T2 and T3 were statistically similar ($P>0.05$) while T1 was the highest and differed significantly ($P<0.05$) from other inclusion levels. The result of the present study is consistent with the findings of Chana et al. (2024). The nitrogen free extract digestibility indicated significant differences ($P<0.05$) across the treatment groups with the digestibility values of 81.73%, 78.16% and 73.92% for T2, T3 and T1 respectively. The higher NFE digestibility values reported in the present study revealed that essential parts of the protein were digested for improving rumen

fermentation which gives considerable balance of nutrient to the animal. Acid detergent fibre digestibility ranged from 56.27 in T1 to 60.27% in T3 respectively without any significant difference ($P>0.05$). The ADF values recorded in this study were comparable with the values reported by Muhammad et al. (2016) for Uda rams fed *Parkia biglobosa*. The NDF value was significantly ($P<0.05$) higher in T3. This implies that fibre fraction digestibility in the 30% inclusion level is better and thus improved microbial activity in the rumen.

Table 5: Nutrient Digestibility of Yankasa Rams Fed Graded Levels of Soybean Husk

Parameters	Treatments			S.E
	T1(0%)	T2(15%)	T3(30%)	
Dry Matter digestibility (%)	75.26 ^a	75.08 ^a	71.35 ^b	0.11
Crude Protein digestibility (%)	79.68	78.68	80.00	2.31
Crude Fibre digestibility (%)	62.56 ^a	50.73 ^b	45.07 ^b	1.95
Ether Extract digestibility (%)	84.39	88.89	88.50	1.69
Nitrogen Free Extract digestibility (%)	73.92 ^a	81.73 ^b	78.16 ^b	0.98
Ash digestibility (%)	80.72	74.94	78.78	1.58
Acid Detergent Fibre digestibility (%)	56.27	59.38	60.27	2.05
Neutral detergent Fibre digestibility	53.91 ^b	53.85 ^b	57.46 ^a	0.71

^{ab}Means with different superscripts across rows are significantly ($p<0.05$) different

CONCLUSION

Based on the findings of this study, soybean husk can be fed to Yankasa breed of sheep up to 30% inclusion level without any harmful effect on the nutrients intake, digestibility and growth performance of the animals.

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